ModelIt!

Teacher's Guide

The Rock That Remembers Everything

How Earth's Rocks Tell the Story of Millions of Years

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-ESS2-1, MS-ESS2-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS2-1, MS-ESS2-3

Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process. Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how heat, pressure, weathering, and time interact to cycle Earth's materials through different rock types.
Disciplinary Core Idea	**ESS2.A Earth Materials and Systems** — All Earth processes are the result of energy flowing and matter cycling within and among the planet's systems, including the rock cycle.
Crosscutting Concept	**Stability and Change** — Students explore how rocks appear stable but are constantly being transformed by slow processes that operate over immense timescales.

CAST Testing Alignment

MULTIPLE CHOICE:

A geologist finds a metamorphic rock that contains distorted fossil shells. Which sequence of events best explains how this rock formed?

CONSTRUCTED RESPONSE:

Explain how a grain of sand on a beach could eventually become part of a metamorphic rock on a mountain. Describe each step of the journey using rock cycle processes.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The rock cycle is one of Earth's most fundamental processes, recycling materials over billions of years. Every rock on Earth has a story to tell about the conditions under which it formed.

Three Rock Types

Igneous rocks form when magma or lava cools and solidifies (granite, basalt, obsidian). Sedimentary rocks form when layers of sediment are compressed and cemented together (limestone, sandstone, shale). Metamorphic rocks form when existing rocks are transformed by heat and pressure without melting (marble from limestone, slate from shale).

The Rock Cycle Process

Rocks continuously transform from one type to another. Igneous rock weathers into sediment, which becomes sedimentary rock. Heat and pressure transform sedimentary rock into metamorphic rock. Melting creates magma that cools into new igneous rock. Any rock type can become any other type — the cycle doesn't follow a fixed order.

Fossils as Evidence

Fossils form when organisms die and are quickly buried by sediment before they decompose. Over time, minerals replace the organic material, preserving the shape. Fossils are only found in sedimentary rock because the heat and pressure that create metamorphic and igneous rocks would destroy them. Fossil distribution provides evidence for continental drift and past environments.

Deep Time

The rock cycle operates on timescales that are difficult to comprehend. The oldest rocks on Earth are about 4 billion years old. A single grain of sand might have been part of a mountain, a beach, a seafloor, and a mountain again over hundreds of millions of years. Understanding deep time is essential for understanding geological processes.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify heat & pressure, weathering rate, sediment layers, and time as the key drivers of the rock cycle.

E — **Establish Relationships**
Students determine that heat & pressure positively drive metamorphism, weathering positively increases sediment production, and time is essential for all transformations.

V — **Visualize & Model**
Students build a model showing how energy inputs and surface processes cycle materials through igneous, sedimentary, and metamorphic stages.

E — **Evaluate Outcomes**
Students run mountain-building and erosion scenarios to observe how different processes dominate at different stages of the cycle.

R — **Revise & Extend**
Students add volcanic activity or biological weathering to explore additional pathways in the rock cycle.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"I'm holding a rock that's older than the dinosaurs. And it has a story to tell."*

Watch For

Pass around fossil specimens or high-quality images. Let students examine them.

Slide 2: Learning Objectives

3 min

Say: *"Today we're reading the autobiography that rocks write — over millions of years."*

Watch For

Have students read objectives. Pre-teach 'sedimentary,' 'metamorphic,' and 'fossil.'

Slide 3: Big Question

3 min

Say: *"How did ocean creatures end up trapped in rocks on the top of Mount Everest?"*

Watch For

Show image of marine fossils found at high elevations. Capture student hypotheses.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the forces that recycle every rock on Earth — over and over for billions of years."*

Watch For

Preview LEVER steps. Connect to prior plate tectonics knowledge from L02.

Slide 5: Activity 1: Components

8 min

Say: *"What forces shape rocks? Which come from inside Earth and which from the surface?"*

Watch For

Guide sorting. Discuss how heat comes from inside Earth while weathering comes from the surface.

Slide 6: Activity 2: Connections

8 min

Say: *"More heat and pressure creates metamorphic rock. More weathering creates more sediment."*

Watch For

Help students trace the cycle: weathering → sediment → compression → metamorphism → melting → igneous.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's build a mountain and then weather it away — over a few hundred million years!"*

Watch For

Students predict outcomes before running. Compare mountain-building vs. erosion scenarios.

Slide 8: Discoveries

5 min

Say: *"Every rock has been recycled for billions of years. The atoms in that rock might have been in a dinosaur!"*

Watch For

Connect to the concept of deep time. Discuss why fossils only survive in sedimentary rock.

Slide 9: STEM Challenge

12 min

Say: *"Can you design something that lasts a MILLION years? Use your rock cycle knowledge!"*

Watch For

Distribute materials. Groups design time capsules using geological principles.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Heat & Pressure, Weathering Rate, Time
- **INTERNAL:** Sediment Layers

Relationships

- **Heat & Pressure to Rock Transformation:** POSITIVE (+) — More heat and pressure drives rock transformation — turning sedimentary into metamorphic rock, or melting rock entirely into magma that becomes igneous rock.
- **Weathering Rate to Sediment Layers:** POSITIVE (+) — Faster weathering breaks more rock into smaller pieces, producing more sediment that accumulates in layers over time.
- **Sediment Layers to Fossil Preservation:** POSITIVE (+) — Thicker, faster-accumulating sediment layers bury organisms more quickly, increasing the chance of fossil preservation before decomposition.

Key Discoveries

- The same atoms that make up a rock today have been recycled through the rock cycle for billions of years
- Extreme heat and pressure transform sedimentary rock into metamorphic rock without melting it
- Fossils are only preserved in sedimentary rock because igneous and metamorphic processes destroy them
- The rock cycle is powered by TWO energy sources: Earth's internal heat and the Sun's energy (driving weathering)

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rocks don't change"	Rocks are constantly changing, just extremely slowly. Over millions of years, mountains erode to plains, seafloors rise to mountaintops, and rocks transform from one type to another. The rock you hold today was once something completely different.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The rock cycle goes in one direction"	Unlike a simple circle, rocks can transform from any type to any other type. Sedimentary rock doesn't HAVE to become metamorphic before becoming igneous — it can melt directly. The 'cycle' is more like a web of possible pathways.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Fossils are just rocks shaped like animals"	Fossils are actual preserved remains where mineral replacement has occurred atom by atom, preserving original structures at microscopic detail. Some fossils even preserve cellular structures, original proteins, and in rare cases, soft tissues.